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Integrating Pulsed-Neutron Interpretation with Reservoir Drive Mechanism in Complex Completion Environment Towards Further Informative Development Plan: A Case Study from Offshore Asia

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Abstract

Pulsed-neutron logging (PNL) interpretation in complex completion wells that aligned with reservoir drive mechanism expectation can be used for further development plan infill well with more confident. Robust workflow is required to adjust the parameter to be align with expected reservoir drive mechanism in dry gas field and edge water drives mechanism. With result from multiple wells, infill well decision for extending field life can be planned confidently.

Interpretation from both inelastic gas mode and sigma modes was performed in multiple wells by comparing result with and without having integrated reservoir knowledge using MCNP model in various completion scenario in the same well (gravel pack, casing only, tubing and casing with various fluid condition). Due to sensitivity of gas mode in this reservoir and wellbore condition, interpretation relies heavily on gas mode. Workflow parameters were adjusted to deal with shaly condition of the reservoir and to be aligned with reservoir mechanism knowledge such as dry gas reservoir gas expansion during depletion or edge water drive reservoir.

Saturation monitoring results from PNL tool reveals dry gas expansion mechanism which gas saturation remains comparable to initial reservoir except one particular direction with significant reduction in water saturation that suggests edge water drive mechanism. This result was based on PNL interpretation from multiple wells in the same field. Robust interpretation workflow from multiple MCNP were used to deal with various completion and fluid condition in tubing and annulus. The saturation result is sensitive to depletion in which accurate knowledge of current reservoir pressure minimize the saturation uncertainty and approach the initial reservoir saturation. Another important factor is Vshale correction in the workflow to deal with shaly sand reservoir that affect the final saturation result. The saturation result was calibrated with current producing section in which remaining gas saturation is still observed. Furthermore, the saturation result is also aligned in the existing perforation with fully water loading. With the result align to both expected reservoir drive mechanism and current production result, the interpretation then can be used as a basis for decision of infill drilling for extending field life.

Multiple wells PNL interpretation for saturation monitoring result that aligned with reservoir drive mechanism and current production results increase the confident in current reservoir fluid condition which leads to more information decision for future infill drilling decision. A collaborative effort between services company and asset teams led to more confident interpretation for future critical field life extension decision of infill drilling in the mature gas field.

Introduction

A prolific gas field lies in the basin in offshore Andaman Sea usually consists of challenges related to sand production from young age reservoir of Pliocene and Pleistocene stratigraphic sequence. The structure of the reservoirs consists of multi-stack of sand shale sequence interpreted to be tidal and fluvial deltaic influenced (Fig. 1) [1]. The producing gas from these reservoirs consists of biogenic and thermogenic nature [2]. The gas accumulated in the three-way dip closure juxtaposed against the faulted. With challenge of producing from such young age reservoir, multiple practices of sand control were implemented to sustain the production [1] [3]. Reservoir monitoring to monitor the gas water contact change for further infill well decision become essential. The challenge of multiple completion type in the well, such as cased-hole gravel pack, single tubing with casing and casing only affects the pulsed neutron response and interpretation. Furthermore, the fluid type in the wellbore and tubing could also affect the pulsed neutron response especially when filled with gas [4]. Analysis without considering the change of the completion could cause misinterpretation or have the interpretation that does not match with reservoir structure (water get into top part of the sand in GP without water effect in the bottom zone).

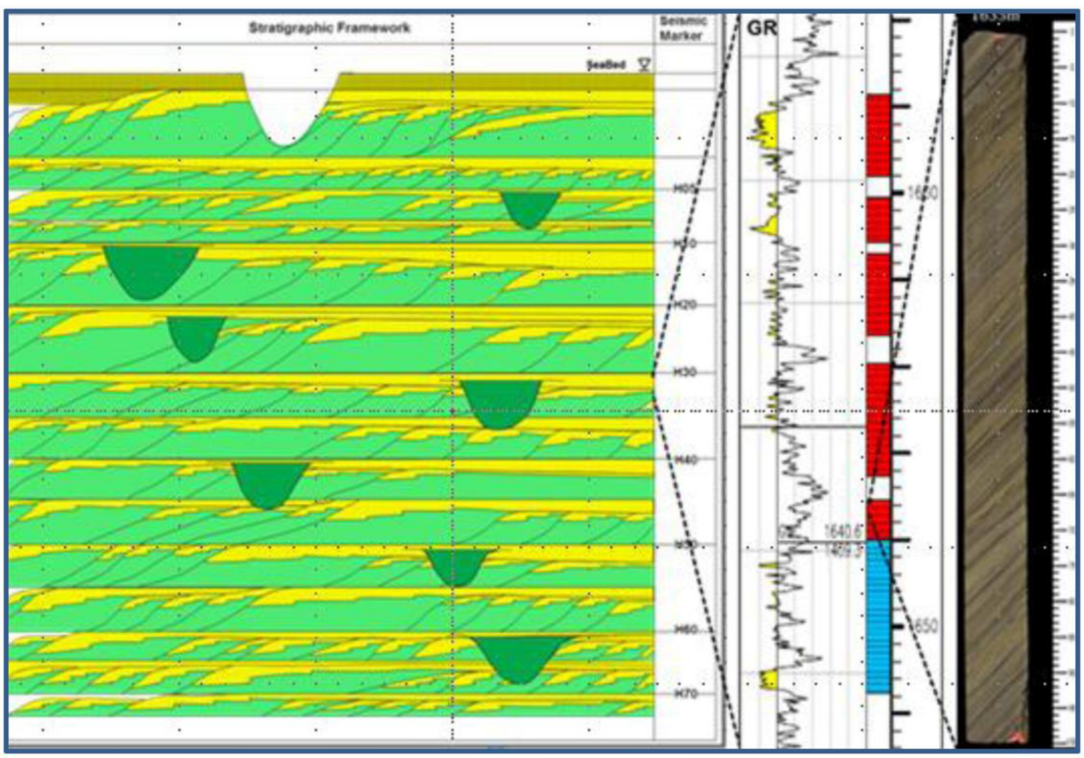

Figure 1—Multiple stacks of reservoir in tidal, fluvial deltaic environment [1]

The interpretation from multiple wells in the same reservoirs could provide more information regarding the contact movement comparison across the field. Single well interpretation may not provide complete picture of the fluid movement. Across the 3 well in the field that PNL were logged, it provides the variation of the GWC trend which then could be related prior knowledge of the reservoir drive mechanism. This correlation of PNL result to reservoir drive mechanism will be discussed.

Methodology

The cased-hole interpretation and petrophysical analysis relies on a well-specific response characterization approach. Each well is modeled individually, considering a range of environmental and completion parameters that significantly influence tool response. The model is developed that accounts for key environmental and completion parameters, including borehole size, casing size and weight, primary lithology (e.g., sandstone, limestone, dolomite), borehole fluid density and salinity, and formation fluid properties such as density, salinity, and gas content. Completion elements such as tubing strings, gravel packs, and annular fluids are also explicitly incorporated. Each configuration was modeled using the same tool geometry to isolate the effects of completion and fluid differences on the detector response. To ensure accurate tool response prediction under these complex and varying conditions, MCNP (Monte Carlo N-Particle) simulations are utilized. This enables a physics-based calibration of the nuclear tool response that captures the influence of both wellbore environment and reservoir characteristics, improving saturation evaluation especially in challenging conditions such as shaly sands, high-pressure gas zones, or mixed fluid environments.

An advanced pulsed neutron modeling was conducted on six distinct zones in Well R using MCNP simulation techniques (Fig. 2). The primary objective of this modeling effort was to accurately evaluate the neutron transport and gamma-ray response in the presence of variable wellbore conditions and fluid environments. This level of detail was essential to ensure reliable formation evaluation across all zones. All six zones in Well R were drilled using an 8.5-inch bit and completed with a 7-inch, 23 pounds per foot (ppf) casing. Despite the consistent outer casing configuration, the tubing type, presence of gravel packs, and fluid content varied between zones, leading to significant differences in neutron behavior and gamma-ray capture response.

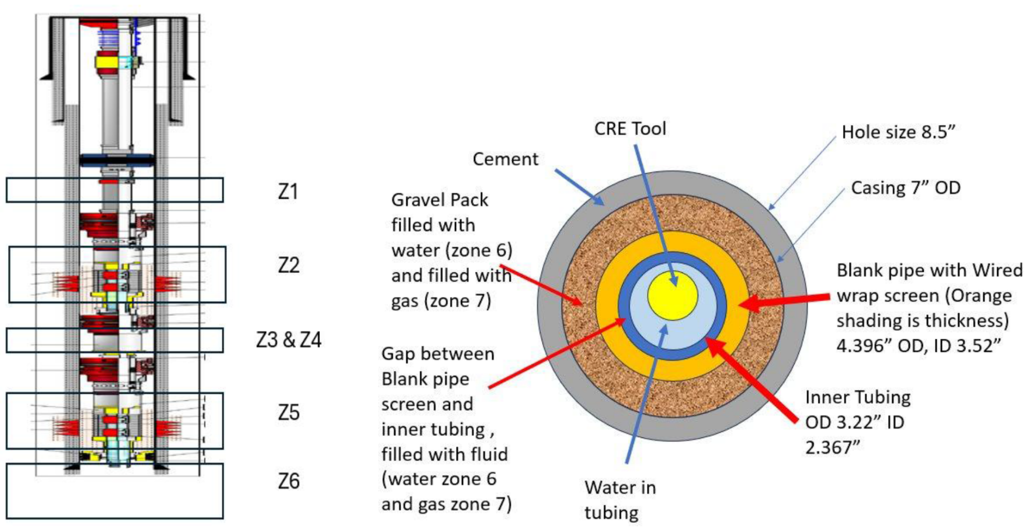

Figure 2—Typical Well Schematic with Cross Section of the Gravel pack zone interval (Cased hole Gravel Pack)

Zone 1 was completed without a gravel pack and utilized a 3.5-inch, gas-filled tubing. The presence of gas had a pronounced effect on the pulsed neutron response, particularly by reducing neutron moderation and thermalization, which then reduce the sensitivity, especially the ratio between the 2 detectors due to neutron transport to be further away in presence of gas. Zone 2 included a cased-hole gravel pack and the same gas-filled tubing as Zone 1. This combination created a unique challenge for the modeling effort.

The gravel pack configuration evaluation possesses uncertainty in terms of fluid type presence in the gravel pack void and gravel pack quality which then lead to more factors to be evaluated even during the modeling. Different MCNP model was generated which show distinct trend from normal tubing and casing configuration (fig. 3). Zones 3 and 4 were completed using 3.5-inch, water-filled tubing, with no gravel pack. The presence of water, having a higher hydrogen index than gas, provided better neutron moderation, resulting in stronger and more interpretable capture ratio response. These zones served as a useful contrast to the gas-filled completions in Zones 1 and 2. Zone 5 incorporated a cased-hole gravel pack similar to Zone 2, but with water-filled tubing. The substitution of gas with water significantly altered neutron behavior. The higher hydrogen content of the water improved thermalization, while the gravel pack still contributed to scattering effects. This combination yielded a neutron transport response that was distinctly different from both fully water-filled and fully gas-filled configurations. Zone 6, the deepest interval in the well, was the least complex in terms of completion. It lacked both tubing and gravel pack, containing only water within the 7-inch casing. This open configuration provided a more straightforward neutron path, but the increased volume of water required careful modeling to properly account for neutron slowing down and capturing events.

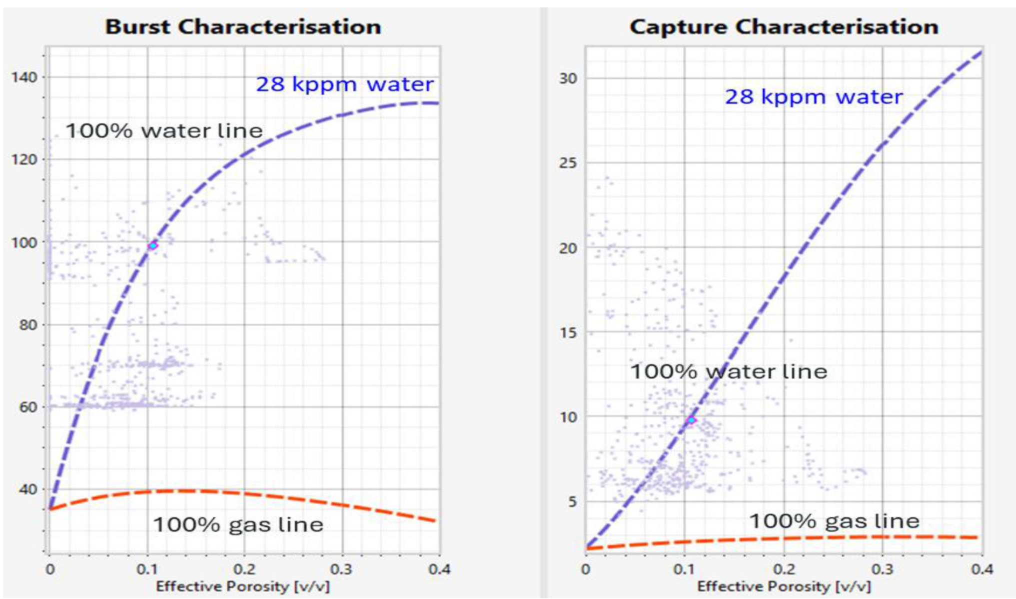

Figure 3—MCNP characterization of the gas mode in the GP completion

For well S, five distinct zones were evaluated using MCNP-based pulsed neutron modeling to capture the varying tool responses under different wellbore conditions and fluid environments. All zones were drilled with an 8.5" bit and completed with a 7" casing and cemented 3.5" tubing. Zones 1 and 3 were completed with cased-hole gravel packs and featured gas-filled tubing, requiring correction for both gravel pack and gas effects, which significantly attenuated capture gamma response and increased reliance on inelastic (gas mode) interpretation. Zones 2 and 4 were without the gravel packs, but while Zone 2 had gas-filled tubing, Zone 4 featured a water-filled tubing, resulting in notably different neutron transport behavior. Zone 5, the deepest, also had water-filled tubing and no gravel pack, allowing for improved sigma mode sensitivity due to higher hydrogen index in the annulus. Across all zones, MCNP simulations were tailored to match each configuration, incorporating tubing fluid, completion hardware, and environmental effects. Water-bearing intervals within each zone served as normalization references, helping to calibrate tool responses and improve saturation estimation. The modeling confirmed that tool response is sensitive to both completion geometry and annular fluid type (Fig. 4), and that zone-specific MCNP calibration is essential for accurate cased-hole interpretation, particularly in gas-bearing and complex completion environments.

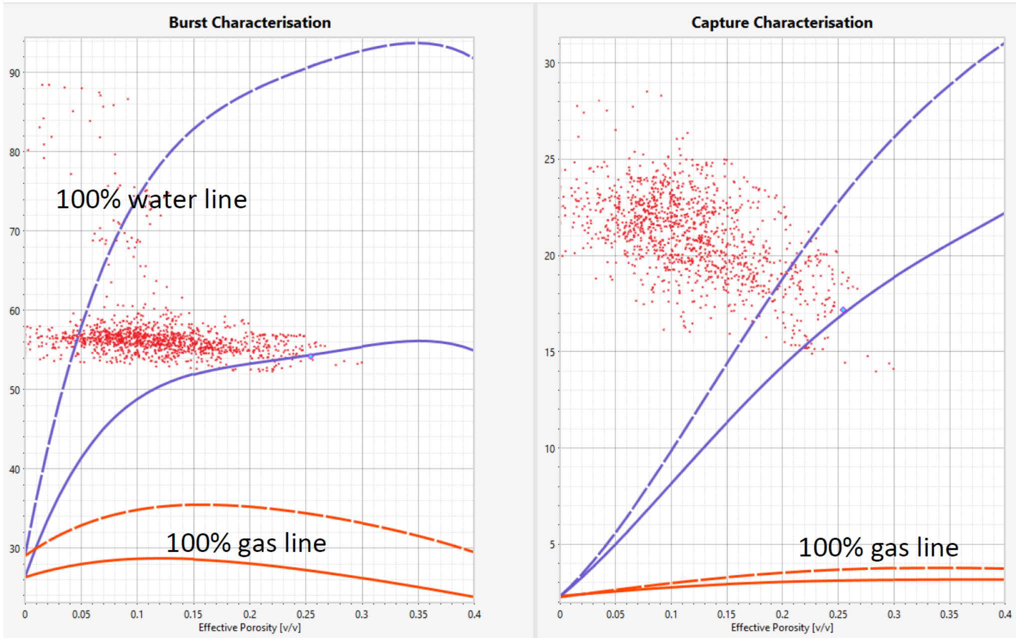

Figure 4—MCNP Characterization of water filled (dash line) and gas filled tubing (solid line) with 7" casing

Overall, the MCNP-based modeling approach proved essential for understanding the complex interplay between well completion design and neutron/gamma responses. These insights are critical for optimizing pulsed neutron logging interpretation and ensuring accurate formation evaluation across diverse wellbore environments.

Interpretation Results – Well R

Zone 1, characterized by gas-filled tubing without a gravel pack, was normalized using a water sand interval at approximately 1800 mMD. This completion configuration inherently reduces neutron moderation due to the presence of gas in the tubing, thereby attenuating the capture gamma response and impacting both burst and capture measurements. The interpretation of this zone revealed a low gas response, with inelastic (burst) mode indicating a gas saturation (Sg) of approximately 10%, while the capture mode exhibited a slightly broader range of 10% to 20% across the analyzed interval (Fig. 5). These saturation values are significantly lower than the open-hole Saturation water effective (SWE) observed between 1906 and 1923 mMD, suggesting the possible fluid movement of water movement to this zone. The weaken hydrogen index, resulting from the gas-filled tubing, likely contributes to the reduced capture response, thus necessitating a greater reliance on inelastic interpretation for a more sensitive estimation of gas saturation.

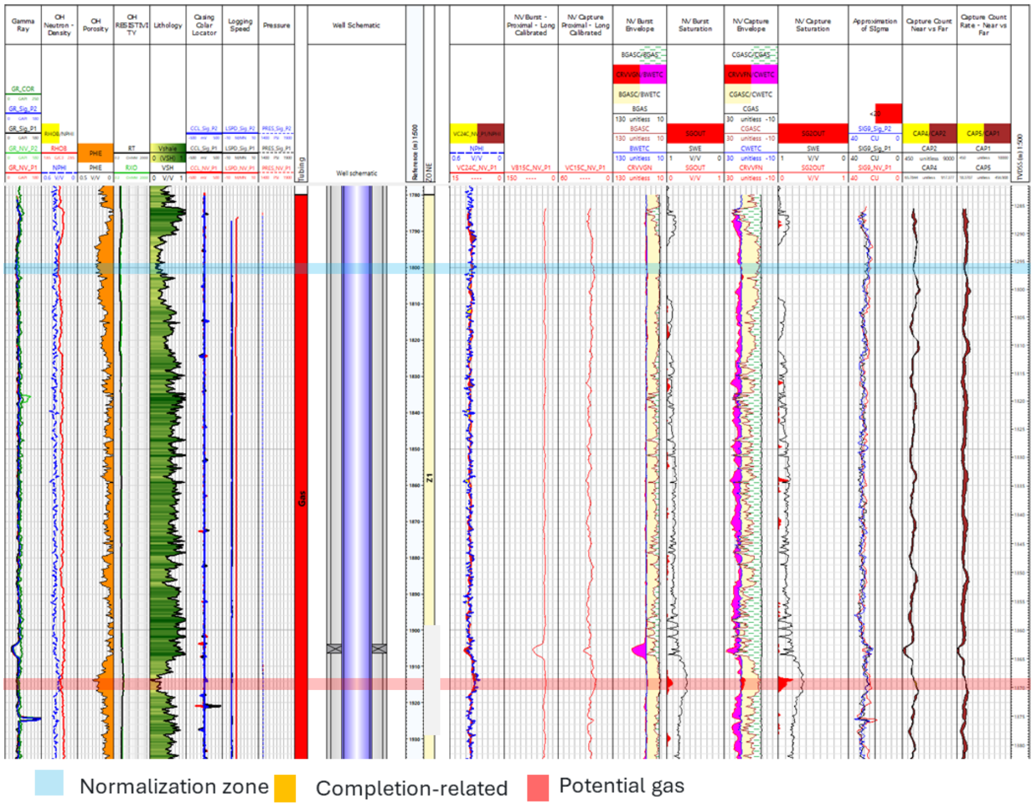

Figure 5—Zone 1 Saturation Result with gas filled tubing inside water filled casing

Zones 2 and 3 consist of intervals completed with gravel packs and gas-filled tubing (Zone 2) as well as water-filled tubing (Zone 3). Water sand normalization was conducted at approximately 2088 mMD. In these intervals, gas responses were clearly observed in both burst and capture modes, with red-shaded log sections identifying the gas-bearing zones. The burst mode yielded gas saturation values ranging from approximately 30% to 55%, while capture mode indicated a broader and higher range between 40% and 80%, particularly in the upper portions of the intervals. Despite the slightly lower saturation values compared to open-hole SWE, the cased-hole interpretations demonstrated a reasonable correlation, suggesting acceptable agreement between cased and open-hole evaluations (Fig.6). An anomalous response was noted in one sand layer, marked in orange, likely resulting from tool interference due to completion equipment, as indicated by a spike in the Casing Collar Locator (CCL) log.

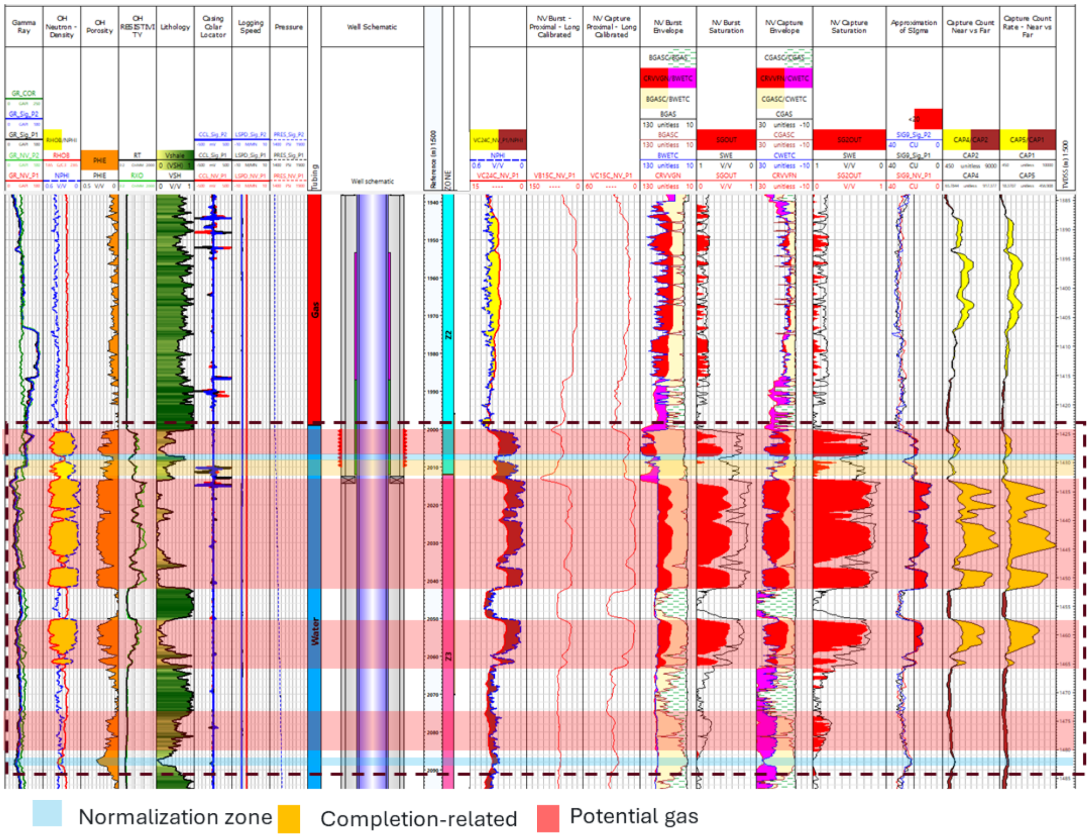

Figure 6—Z2 and Z3 for gas filled and water filled tubing in CHGP completion

Zones 5 and 6 present a contrast to the previous completions, as they are completed with water-filled environments. Zone 5 are located in another reservoir layer, includes a gravel pack, whereas Zone 6 comprises a 7-inch water-filled casing without tubing or gravel pack. Both zones are spanned by a single casing section. In Zone 5, the upper section above 2245 mMD is identified as a shale interval, underlain by reservoir sands. Water sand normalization was applied at 2247.5 mMD for Zone 5 and at 2377.8 mMD for Zone 6. Distinct gas responses were evident in the red-highlighted intervals of both zones. Inelastic interpretation revealed gas saturation ranging from approximately 15% to 30%, while capture mode showed stronger responses, ranging from 30% to 60% Sg (Fig. 7). This behavior is consistent with expectations, as the water-filled tubing enhances hydrogen index sensitivity, thereby improving the capture gamma response in the absence of attenuation from gas-filled completions. Despite these enhanced responses, the saturation values remained significantly lower than those recorded in the open-hole SWE, which may indicate either partial gas filling or underestimation of saturation due to local reservoir heterogeneity. Based on the observed shift in saturation trends, a GDT was interpreted at approximately 2302 mMDDF, aligning well with formation characteristics and log-derived data.

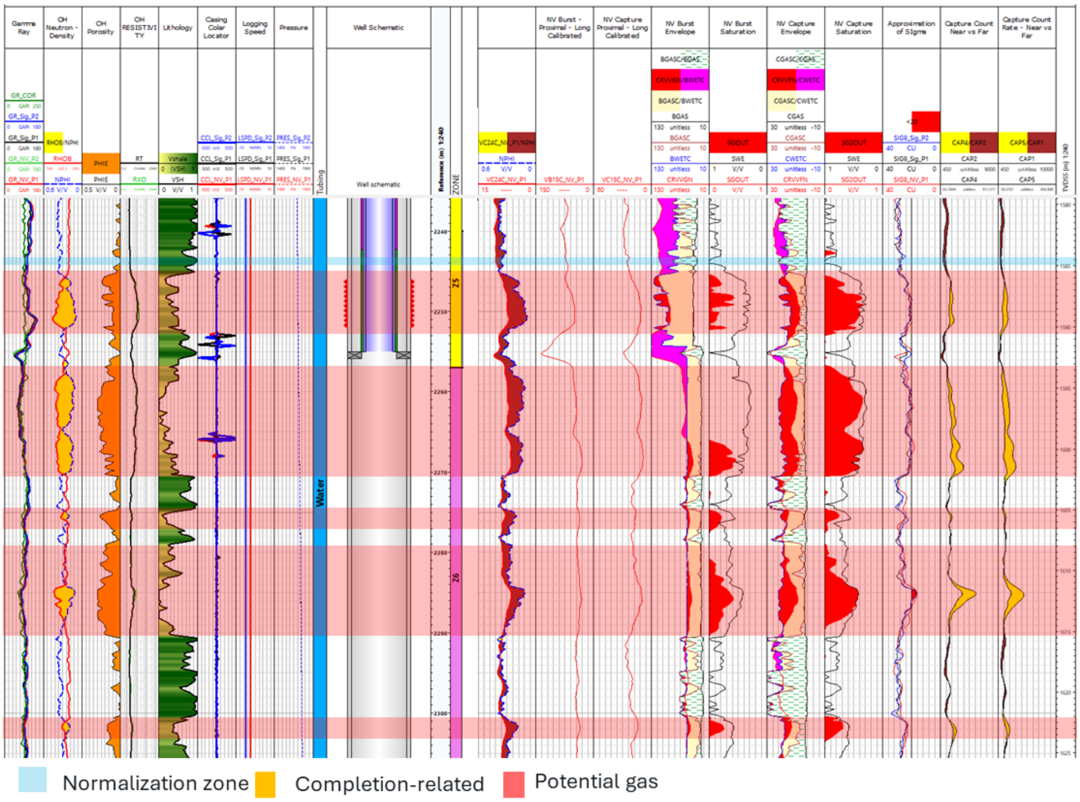

Figure 7—Z5 & Z6 Saturation result in CHGP and below sump packer

Interpretation Results – Well S

The pulse neutron interpretation and analysis for Well S intersects two major reservoirs, spanning five evaluation zones (Zones 1–5), each with distinct fluid content, lithology, and completion designs. Across these intervals, MCNP-assisted pulsed neutron interpretation was performed using both burst (inelastic) and capture (sigma) modes, with environmental corrections based on tubing fluid, gravel pack, and casing effects. Results reveal mode-dependent saturation sensitivity and completion-driven variations in tool response, as well as depletion behavior in gas-bearing intervals.

In Zone 1, the interval is completed with a single casing, gravel pack, and gas-filled tubing. Shale normalization was performed at ~1xxx.9 m MDDF. Both burst (inelastic) and capture (sigma) modes detected clear gas signatures, but capture mode provided more reliable results, showing 80–90% gas saturation (Sg) compared to 50–70% from burst. Capture saturation aligned well with openhole SWE logs, while burst saturation underestimated, likely due to attenuation from gas-filled tubing and gravel pack (Fig 8). A possible Gas Down To (GDT) was interpreted at ~1xxx m MDDF, suggesting a depletion toward the base of the interval.

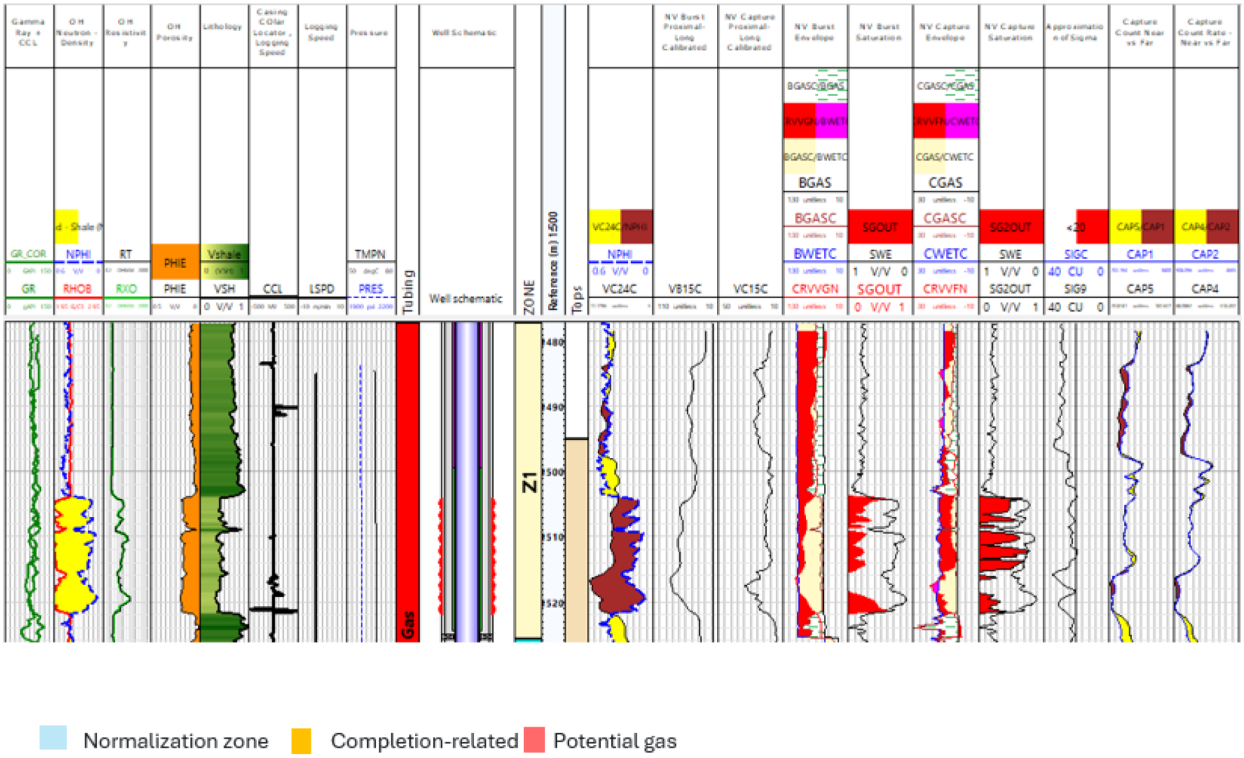

Figure 8—Z1 Saturation result across GP with gas filled tubing

In Zone 2, the interval is characterized by a shale-dominated formation with gas-filled tubing, and no gravel pack. A water sand normalization point was defined at ~1xxx m MDDF, based on completion and log data. In this zone, no significant gas response was observed in either burst or capture modes, suggesting a combination of poor reservoir quality and possibly depletion (Fig.9). The lack of response, despite gas in the tubing, indicates that tool counts are heavily influenced by the surrounding formation lithology and not just fluid content within the borehole. This reinforces the need for proper environmental correction using normalization intervals in shale zones.

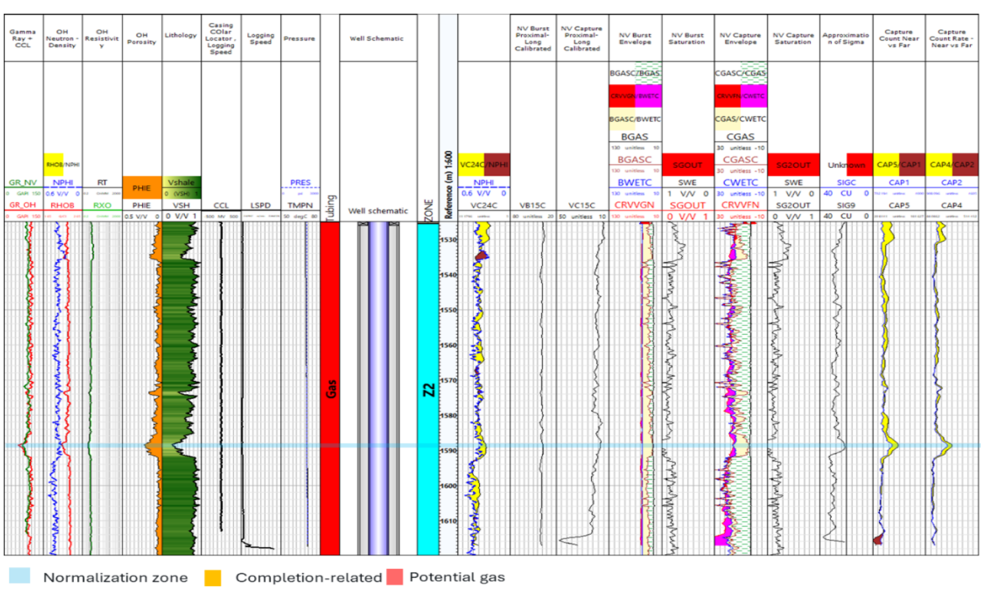

Figure 9—Z2 Saturation result with gas filled tubing and water filled annulus

In contrast for Zone 3, although completed with the same hardware and gas-filled tubing, it is a shale-dominated interval with lower reservoir quality. A shale normalization point was set at ~1xxx.x5 m MDDF. In this zone, no significant gas signature was detected in either burst or capture mode. The absence of response despite similar completion design indicates a non-reservoir interval which is characterized as a predominantly shale interval reinforcing the critical role of formation properties and prior production in gas detectability. MCNP modeling confirmed that any minor variations observed in the raw counts were attributed to environmental effects rather than fluid content changes, reaffirming the value of normalization using shale baseline (Fig. 10)

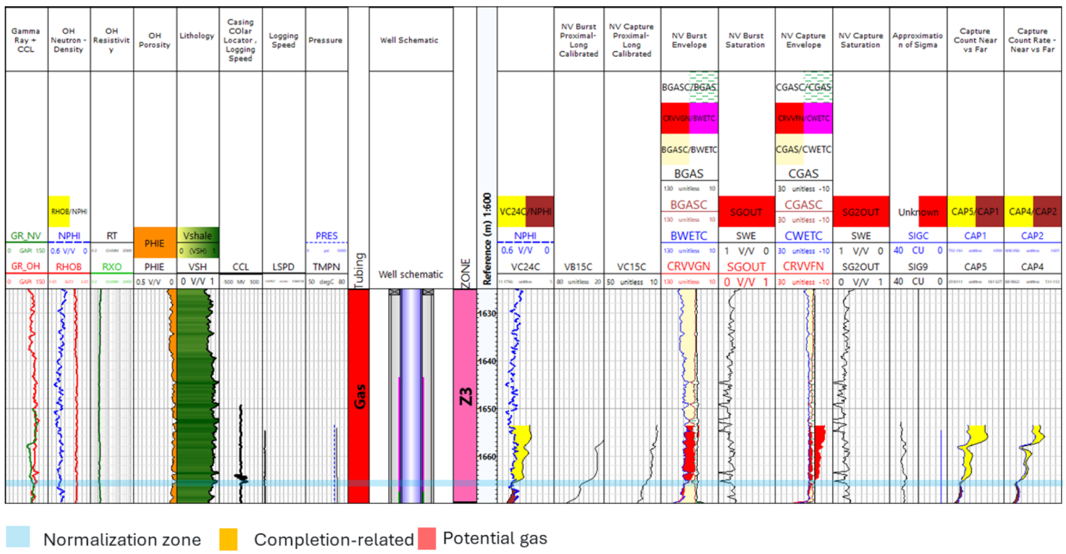

Figure 10—Z3 Saturation result with gas filled tubing and water filled annulus within GP completion

A key completion change occurs in Zone 4 where the tubing fluid changes from gas to water, while gravel pack remains, Zone 4 displayed a clear increase in capture mode sensitivity due to the shift to water-filled tubing, which enhances neutron moderation in the near-borehole region. A shale normalization point was established at ~1XXX.4 m MDDF, serving as a baseline for gas saturation calculations. In this zone, a clear gas response was observed in both burst and capture saturation profiles, with capture saturation indicating ~80–90% Sg and burst saturation ranging between 40–50% Sg (Fig. 11). The observed difference emphasizes the stronger signal strength and reliability of the capture mode in water-filled completions. The lower burst values are expected due to residual attenuation by the gravel pack and formation shaliness.

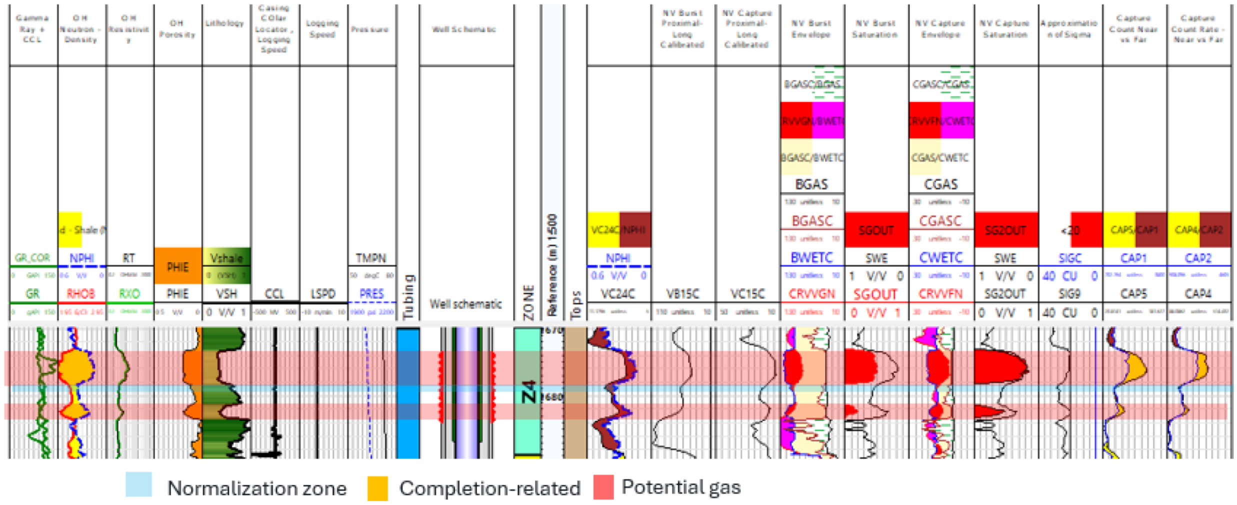

Figure 11—Z4 saturation result across CHGP with water filled tubing

Zone 5 (1xxx–1xxx m MDDF) in Well S represents the deepest analyzed section, completed with single casing, water-filled tubing, and no gravel pack. This well configuration provides an ideal environment for pulsed neutron tools, minimizing attenuation and enhancing both burst and capture responses. A water sand normalization point was established at ~1xxx m MDDF to anchor saturation interpretations. Despite the favorable tool environment, this zone showed only thin and discontinuous gas-bearing intervals, suggesting depletion effects or limited reservoir development. Two distinct gas responses were identified. The first occurred between 1A-1B7m MDDF, where burst saturation reached 40–50% Sg and capture saturation was slightly higher at 50–60% Sg, indicating a modest but clear gas-bearing zone. The second response, from 1710.40–1712.00 m MDDF, was weaker and thinner, with both burst and capture indicating only 10–20% Sg (Fig. 12). These subtle gas signatures particularly were more clearly defined in capture mode, which remained consistent with SWE from open hole evaluation, while burst underrepresented saturation due to the reduced inelastic interaction in low-mobility or thin gas zones.

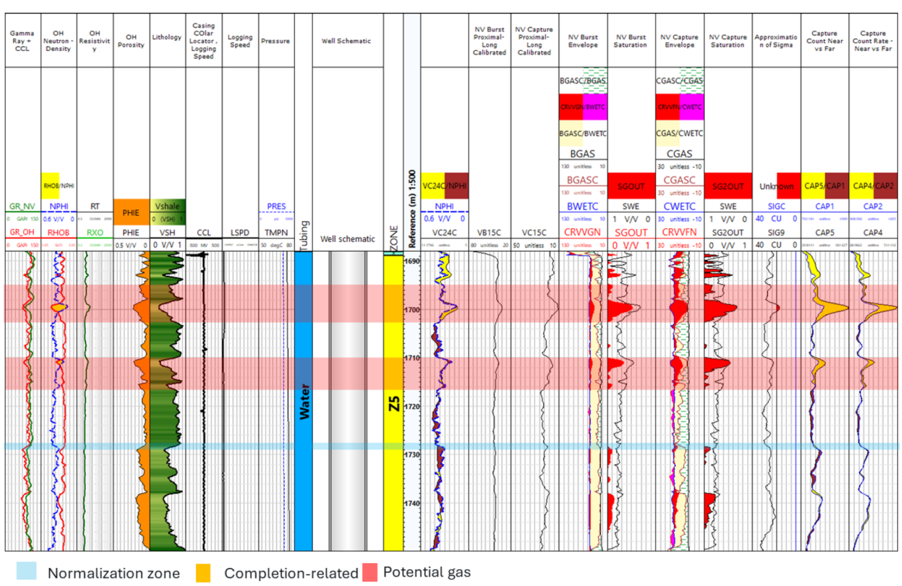

Figure 12—Z5 Saturation Result below the sump packer

The absence of gravel pack and the presence of water-filled tubing significantly improved neutron moderation and gamma-ray production, which in turn enhanced the sigma (capture) measurement's sensitivity to gas. This allowed reliable identification of gas zones even in thin-bedded intervals. The detection of a probable GDT at ~1xxx m MDDF indicates the base of mobile gas, possibly linked to depletion from overlying production zones or lateral drainage. These findings further validate the need for MCNP-assisted modeling and completion-sensitive interpretation workflows, especially in deep, transitional reservoir intervals where gas responses may be subtle and discontinuous.

MULTIWELL CORRELATION RESULT

The multi-well correlation plot from the 3 wells suggests the well towards one side of the reservoir still shows significant gas saturation. While on another side of the reservoir, significant gas saturation reduction is observed. With assumption of depletion drive of gas reservoir that gas still expand that the saturation after depletion, the saturation trend to match with original gas saturation on a few wells on the side of the reservoir. While on another side of the reservoir, from the processing and interpretation, there is significant gas saturation reduction that likely suggests water encroachment. From interpretation variation in term of parameter adjustment, the variation in the gas saturation results still see significant gas reduction that indicates the water encroachment especially on one side of the reservoir.

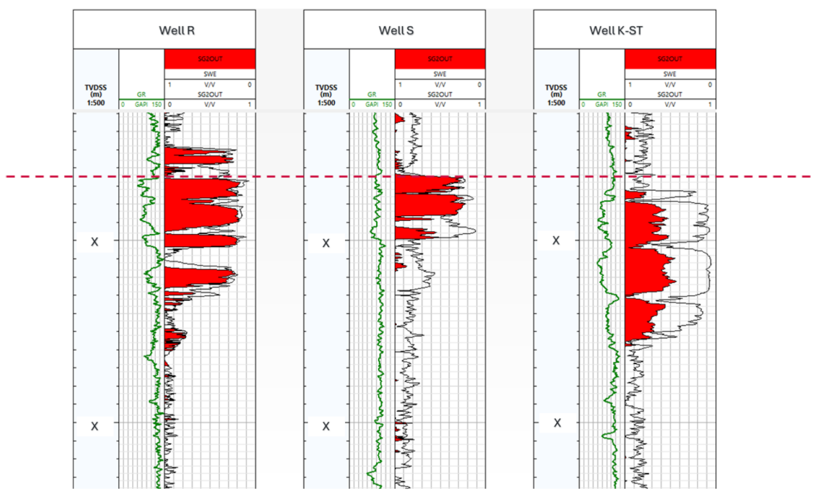

Figure 13—Multi-well correlation from the 3 well prior to incorporate reservoir structure knowledge

By combining with existing reservoir drive mechanism, the edge water drive reservoir mechanism is expected. This seems to be in alignment with PNL interpretation result in which the well located on the east side of the reservoir is expected to have more edge water drive than the other. This is also confirmed with current production that the well on the west side is still an active producer while the well on the east is fully loaded. By having the interpretation aligned with the expectation from reservoir drive mechanism, it increases the confidence of the interpretation and hence able to plan the next infill drilling well.

The multi-well comparison plot (depth in TVDss) shows variation of remaining gas saturation from the 3 wells across the structure of one reservoir layer. The dash line refers to the same TVDss reference. From this reference datum, GWC could be determined from the first 2 well (from the left) where the trend of significant saturation change is observed. Where else the well on the right (east side), no clear GWC is observed but rather water encroachment from base to top of the sand. Hence there is no single GWC determined from the current interpretation but rather tilted contact in which on the east side there is more effect from edge water drive. This was also aligned with dynamic reservoir expectation of edge water drive.

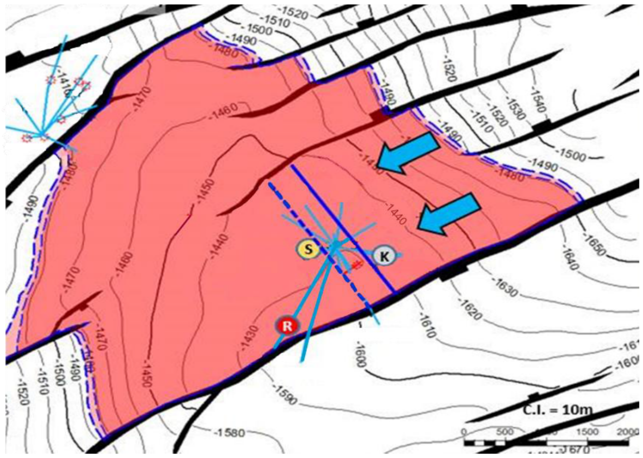

Figure 14—Reservoir top structure with three-way dip closure juxtaposed against fault with blue arrow highlighting expected edge water drive

By combining reservoir drive mechanism together with PNL interpretation, this can be used to justify the infill well location that extends the gas field life production and delays the water breakthrough of the new infill well. By having the PNL interpretation result that aligned with reservoir drive mechanism, it could increase confidence in the planning of the infill well.

Conclusion

PNL interpretation results from multiple wells across the reservoir structure can be incorporated together with the expected reservoir drive mechanism knowledge for more confident reservoir surveillance and future production. Furthermore, this interpretation is also aligned with current production in which some of the well is fully loaded and some are still active producers. With these complex completion changes, the interpretation workflow needs to handle the borehole configuration change. One of the ways to verify is the consistent result even across the completion change. With result aligned across the multiple well with the reservoir drive mechanism, a collaborative effort between services company and asset teams led to more confident interpretation for future critical field life extension decision of infill drilling in the mature gas field is achieved.

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